



# DOCKER CHEAT SHEET — ESSENTIAL COMMANDS (Compact)

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## ▶ VERSION & INFO

- `docker version` — client/server version
- `docker version --format '{{.Server.Version}}'`
- `docker info` — system-wide info
- `docker system df` — disk usage
- `docker stats` / `--all` — live resource usage

## ▶ CONTAINERS — LIFECYCLE

|                                             |                             |
|---------------------------------------------|-----------------------------|
| <code>docker create image</code>            | create (not start)          |
| <code>docker run image</code>               | create + start              |
| <code>docker run -td img</code>             | detached + tty (keep alive) |
| <code>docker run --rm img</code>            | auto-remove on exit         |
| <code>docker run --name myapp img</code>    | name container              |
| <code>docker rename old new</code>          | rename                      |
| <code>docker rm ctr</code>                  | delete stopped              |
| <code>docker rm -v ctr</code>               | + remove volumes            |
| <code>docker rm -f \$(docker ps -qa)</code> | delete ALL (force)          |
| <code>docker update ctr</code>              | update limits               |

### Run Options

- `-d` detached, `-t` tty, `-i` interactive → `-it`
- `-p 127.0.0.1:HOST:CTR` port map
- `-v $HOST:$CTR` bind mount
- `-v vol:/path:ro` read-only vol
- `--volumes-from ctr` share volumes
- `-e VAR=val` env var, `--env-file`
- `--log-driver=syslog` custom log

## ▶ CONTAINERS — START / STOP

|                                       |                     |
|---------------------------------------|---------------------|
| <code>docker start ctr</code>         | start stopped       |
| <code>docker stop ctr</code>          | graceful stop       |
| <code>docker restart ctr</code>       | stop + start        |
| <code>docker pause / unpause</code>   | freeze / unfreeze   |
| <code>docker wait ctr</code>          | block until stop    |
| <code>docker kill ctr</code>          | SIGKILL             |
| <code>docker attach ctr</code>        | attach console      |
| <code>Ctrl+p, Ctrl+q</code>           | detach without stop |
| <code>docker exec -it ctr bash</code> | new shell inside    |

### CPU / Memory

- `docker run -c 512 img` — 50% CPU (1024=100%)
- `docker run --cpuset-cpus=0,4,6` — pin cores
- `docker run -m 300M ubuntu:14.04 bash` — mem limit
- `--device=/dev/ttyUSB0 / --privileged -v /dev/bus/usb:/dev/bus/usb`
- `--cap-add SYS_ADMIN --device /dev/fuse` — FUSE

## ▶ CONTAINERS — INFO

|                                                                     |                      |
|---------------------------------------------------------------------|----------------------|
| <code>docker ps</code>                                              | running only         |
| <code>docker ps -a</code>                                           | all (incl. stopped)  |
| <code>docker ps -l -q</code>                                        | last container ID    |
| <code>docker logs ctr</code>                                        | logs (json/journald) |
| <code>docker inspect ctr</code>                                     | full JSON (IP etc)   |
| <code>docker inspect -f '{{.NetworkSettings.IPAddress}}' ctr</code> | template             |
| <code>docker port ctr</code>                                        | port mapping         |
| <code>docker top ctr</code>                                         | processes            |
| <code>docker diff ctr</code>                                        | changed files        |
| <code>docker events</code>                                          | live events          |
| <code>docker cp ctr:/path ./host</code>                             | copy out/in          |
| <code>docker export ctr   gzip &gt; ctr.tar.gz</code>               | fs to tar            |

### Quick IP / Port

- `docker inspect $(dl) | grep IPAddress | cut -d'"' -f4`
- `docker inspect -f '{{range $p, $c := .NetworkSettings.Ports}} {{p}} -> {{{index $c 0}.HostPort}} {{end}}' ctr`

## ▶ IMAGES

|                                                               |                 |
|---------------------------------------------------------------|-----------------|
| <code>docker images</code>                                    | list images     |
| <code>docker images -viz   dot -Tpng -o img.png</code>        | deps graph      |
| <code>docker build -t name:tag .</code>                       | from Dockerfile |
| <code>docker build --build-arg KEY=val .</code>               | build args      |
| <code>docker commit ctr img:tag</code>                        | ctr → image     |
| <code>docker tag src dst</code>                               | tag image       |
| <code>docker history img</code>                               | layer history   |
| <code>docker rmi img</code>                                   | remove image    |
| <code>docker rmi \$(docker images -q -f dangling=true)</code> | dangling        |
| <code>docker rmi \$(docker images -q)</code>                  | ALL images      |

### Save / Load vs Import / Export

- `docker save myimg:tag | gzip > img.tar.gz` — with history
- `docker load < img.tar.gz` — load with history
- `docker export myctr | gzip > ctr.tar.gz` — fs only, no history
- `cat ctr.tar.gz | docker import - myimg:tag` — smaller

## ▶ NETWORKS

- `default: bridge, host, none` container → `bridge` by default

|                                                                                             |                |
|---------------------------------------------------------------------------------------------|----------------|
| <code>docker network create net</code>                                                      | bridge default |
| <code>docker network create --subnet 203.0.113.0/24 --gateway 203.0.113.254 iptastic</code> | custom subnet  |
| <code>docker run --net iptastic -ip 203.0.113.2 nginx</code>                                | static IP      |
| <code>docker network ls</code>                                                              | list           |
| <code>docker network inspect net</code>                                                     | detail         |
| <code>docker network connect net ctr</code>                                                 | attach ctr     |
| <code>docker network disconnect net ctr</code>                                              | detach         |
| <code>docker network rm net</code>                                                          | remove         |

## ▶ VOLUMES

|                                                               |                |
|---------------------------------------------------------------|----------------|
| <code>docker volume create vol</code>                         | create         |
| <code>docker volume ls</code>                                 | list           |
| <code>docker volume inspect vol</code>                        | detail         |
| <code>docker volume rm vol</code>                             | remove         |
| <code>docker run -v /host:/ctr</code>                         | bind host dir  |
| <code>docker run -v vol:/data</code>                          | named volume   |
| <code>docker run -v \$PWD/conf:/conf:ro -p 80:80 httpd</code> | file as volume |

- `docker run --volumes-from ctr` — mount from data-ctr (legacy, now named vol)

- Named volumes replace data-only containers since 1.9.0

## ▶ REGISTRY & REPOSITORY

|                                               |                |
|-----------------------------------------------|----------------|
| <code>docker login</code>                     | login          |
| <code>docker logout</code>                    | logout         |
| <code>docker search term</code>               | search Hub     |
| <code>docker pull img:tag</code>              | pull           |
| <code>docker pull debian@sha256:abc...</code> | verify by hash |
| <code>docker push img:tag</code>              | push           |

- Local registry: `docker/distribution` — see local deploy docs

## ▶ DOCKERFILE — INSTRUCTIONS

|                                                                  |                                     |
|------------------------------------------------------------------|-------------------------------------|
| <code>FROM img</code>                                            | base image                          |
| <code>RUN cmd</code>                                             | exec + commit layer                 |
| <code>CMD ["exe", "param"]</code>                                | default cmd                         |
| <code>ENTRYPOINT ["exe"]</code>                                  | as executable                       |
| <code>EXPOSE 80</code>                                           | declare port (needs -p)             |
| <code>ENV KEY=val</code>                                         | env var                             |
| <code>ARG KEY=val</code>                                         | build-time var                      |
| <code>COPY src dst [--chown=user:grp]</code>                     | copy files                          |
| <code>ADD src dst</code>                                         | copy + remote/tar (avoid, use COPY) |
| <code>VOLUME /path</code>                                        | mount point                         |
| <code>USER user</code>                                           | next cmds user                      |
| <code>WORKDIR /path</code>                                       | workdir                             |
| <code>LABEL key=val</code>                                       | metadata                            |
| <code>ONBUILD instr</code>                                       | trigger for child build             |
| <code>STOPSIGNAL SIGTERM</code>                                  | stop signal                         |
| <code>SHELL ["/bin/sh", "-c"]</code>                             | override shell                      |
| <code>HEALTHCHECK CMD curl -f http://localhost/    exit 1</code> | health check                        |

- `.dockerignore` — exclude files from context
- `RUN groupadd -r u && useradd -r -g u u; USER u` — non-root

## ▶ PORTS — EXPOSING

- `docker run -p 127.0.0.1:HOST:CTR -t img` — host→container
- `EXPOSE <PORT>` — doc only, -p does expose
- `docker port CTR CTRPORT` — show mapping
- `iptables -t nat -A DOCKER -p tcp --dport HOST -j DNAT --to-destination CTR_IP:PORT`
- Vagrant: `(49000..49900).each {|p| config.vm.network :forwarded_port, host: p, guest: p}`

## ▶ DOCKER COMPOSE

- `docker-compose -f file.yml up` — start multi-ctr app
- `docker-compose up -d` — detached
- `docker-compose stop` — stop
- `docker-compose down` — stop+rm ctr/net
- `docker-compose down --volumes` — + vols
- v2: `docker compose up` — (no hyphen)

## ▶ CLEANUP — PRUNE & SYSTEM

|                                                                      |               |
|----------------------------------------------------------------------|---------------|
| <code>docker system prune</code>                                     | all unused    |
| <code>docker container prune</code>                                  | stopped ctrs  |
| <code>docker image prune</code>                                      | dangling imgs |
| <code>docker volume prune</code>                                     | dangling vols |
| <code>docker network prune</code>                                    | unused nets   |
| <code>docker rm -v \$(docker ps -aq -f status=exited)</code>         | exited ctrs   |
| <code>docker volume rm \$(docker volume ls -qf dangling=true)</code> | dangling vols |

### One-liners

- `docker kill $(docker ps -q)` — kill running
- `docker stop $(docker ps -aq) && docker rm -v $(docker ps -aq)`
- `docker ps -a | grep 'weeks ago' | awk '{print $1}' | xargs docker rm`
- `docker ps -a | grep Exit | awk '{print $1}' | xargs docker rm -f`

► TIPS — COMPACT

- **Heredoc build:** `docker build -t http - << EOF FROM alpine  
RUN apk add http EOF`
- **Last ID:** `alias dl='docker ps -l -q'; docker commit $(dl) helloworld`
- **Commit custom cmd:** `docker commit -run='{ "Cmd":  
["postgres", "-too-many-opts"]}' $(dl) postgres`
- **Env:** `docker run --rm ubuntu env`
- **Monitor:** `docker stats $(docker ps -q) / docker stats  
$(docker ps --format '{{.Names}}')`
- **Find by regex:** `for i in $(docker ps -a  
| grep REGEX  
| cut -f1 -d" ");  
do echo $i; done`
- **Slim:** `RUN apt cmds && apt-get clean && rm -rf  
/var/lib/apt/lists/* /tmp/*`
- **Flatten:** `ID=$(docker run -d img bash);  
docker export $ID  
| docker import - flat-img`

► SECURITY — ESSENTIALS

- `Δ docker = root` group `docker`  $\approx$  root
- `docker run --read-only` — read-only fs
- `docker run -v $(pwd)/secrets:/secrets:ro` — ro vol
- `docker run --pids-limit=64` — prevent fork bomb ( $k \geq 4.3$ )
- `docker run --security-opt=no-new-privileges`
- `docker -d --icc=false --iptables` — no inter-ctr comm
- `USER user` in Dockerfile — non-root
- `docker pull debian@sha256:...` — verified pull
- Run Docker in VM + AppArmor / SELinux / seccomp
- Bench: `docker/docker-bench-security`

► INSTALLATION — QUICK

- **Linux:** `curl -sSL https://get.docker.com/ | sh`
- **macOS:** `brew install --cask docker & docker run hello-world`
- **Win:** Docker Desktop → `docker version` → `docker run hello-world`
- **Check:** `docker version --format '{{.Server.Version}}'`

Layers / Links (legacy)

- Layers = versioned fs  $\approx$  git commits
- `docker run -d --link CONTAINER:ALIAS --name LINKED user/wordpress` → `ALIAS_PORT_1337_TCP_ADDR` (deprecated → user-defined networks)